

(2) [3+3+3+3=12 points]

Carefully consider the following applications, and indicate which ADT is most appropriate for each (map, set, list, stack, or queue). Make sure to explain your reasoning. More detail is better: instead of just "List" you could state "a list of student records".

- (a) Data structure allowing you to find a book's title from its ISBN identifier (13 characters, mostly digits, but could contain the letter 'X').

map < ISBN, title >
string/int string

- (b) Waitlist of students who want to enroll in CS104, but couldn't get in yet.

Queue < student >
string? → maintains order & is the most restrictive/specific ADT that is possible

- (c) Data structure to store the content of each line of code of a possibly long C++ program.

List < ~~the~~ line of code >
string → order is important... thus a list

- (d) A data structure that allows you to input an academic semester and find all students who earned an A in CS104 that semester.

map < semester, set < student > >
string/int string?

→ set allows efficient membership/find operation